

PROJECT REPORT

CUSTOMER CHURN ANALYSIS

An Exploratory Data Analysis Using Python

PREPARED BY

JUMANA

Python | Pandas | NumPy

Matplotlib | Seaborn

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Executive Summary

This project explores customer churn in a telecommunications dataset using Python. The analysis focuses on customer complaints, service usage, customer value, age, tariff plans and other behavioural indicators to identify patterns associated with customers leaving the service.

After removing 300 duplicate records, the cleaned dataset contained 2,850 customer records. Of these, 446 customers (15.65%) had churned, while 2,404 (84.35%) had remained. The strongest observed pattern was the relationship between complaints and churn: 82.61% of customers who complained had churned, compared with 9.77% of customers who did not complain.

Churned customers also showed substantially lower average call usage, SMS usage and customer value than retained customers. These findings can help a telecom business prioritise complaint resolution, monitor declining engagement and investigate customer groups with higher observed churn.

1. Introduction

Customer churn occurs when customers stop using a company's products or services. In the telecommunications industry, churn can affect revenue, customer growth and long-term business performance. Understanding the patterns associated with churn can help companies improve customer experience and retention.

This project uses exploratory data analysis (EDA) to examine customer information and service usage. Python libraries are used to clean, summarise, analyse and visualise the dataset. The work is descriptive and exploratory; it does not build a machine-learning churn prediction model.

2. Problem Statement

Telecommunication companies may lose customers when service experiences, usage patterns or customer needs are not adequately addressed. The business problem considered in this project is to understand which customer characteristics and behaviours show notable differences between customers who churned and those who stayed.

3. Project Objectives

- Measure the overall customer churn rate.
- Explore customer complaints and their association with churn.
- Compare call and SMS usage between churned and retained customers.
- Examine customer value across churn groups.
- Explore churn patterns across age and tariff-plan categories.
- Use visualisations and correlation analysis to communicate key patterns.
- Develop practical business recommendations from the findings.

4. Dataset Description

The dataset used for this analysis contains telecommunications customer records and behavioural attributes. The original dataset contained 3,150 rows and 14 columns. After exact duplicate records were removed, 2,850 rows remained for analysis.

Dataset detail	Description
Original size	3,150 rows × 14 columns
Records removed	300 duplicate rows
Cleaned size	2,850 rows × 14 columns
Missing values	No missing values were observed in the initial inspection

Target field	Churn (0 = stayed, 1 = churned)
Key fields	Complains, Seconds of Use, Frequency of use, Frequency of SMS, Age, Tariff Plan, Customer Value

Note: Age Group, Tariff Plan and Status are represented by numeric codes in the dataset. The analysis treats these as coded categories and does not assign unverified descriptive labels to the codes.

Dataset Source:

The dataset used for this project was obtained from the UCI Machine Learning Repository.

Source: https://archive.ics.uci.edu/dataset/563/iranian+churn+dataset

5. Tools and Technologies

Tool / Library	Purpose
Python	Programming language used for the analysis
Pandas	Data loading, cleaning, grouping and summarisation
NumPy	Numerical operations
Matplotlib	Charts and plots
Seaborn	Statistical visualisation
Jupyter Notebook	Interactive environment for analysis and documentation

6. Methodology

6.1 Data Exploration

The dataset was inspected to understand its dimensions, column names, data types, non-null counts and descriptive statistics. This step provided an overview of the available customer and usage variables.

6.2 Data Cleaning

Duplicate records were identified and removed. The dataset decreased from 3,150 rows to 2,850 rows. A subsequent duplicate check returned zero duplicate rows. The initial inspection found no missing values.

6.3 Exploratory Analysis

The analysis compared churn across complaint status, age, tariff plan and status categories. It also compared average service usage and customer value between churn groups.

6.4 Visualisation and Correlation

Bar charts, scatter plots, a customer-value distribution plot and a correlation heatmap were used to explore patterns and relationships among selected variables. Correlation values describe association and should not be interpreted as evidence of causation.

7. Exploratory Data Analysis and Results

7.1 Overall Customer Churn

Churn status	Number of customers	Percentage

Stayed (0)	2,404	84.35%
Churned (1)	446	15.65%
Total	2,850	100.00%

The overall churn rate was 15.65%. Most customers in the cleaned dataset remained with the telecom company.

7.2 Complaints and Customer Churn

Complaint status	Stayed	Churned	Churn rate
No complaint (0)	2,364	256	9.77%
Complaint (1)	40	190	82.61%

Customers who recorded a complaint had a much higher observed churn rate than customers without a complaint. This is a strong pattern in the analysed data and suggests that complaint handling deserves close attention. The result does not by itself prove that a complaint caused a customer to leave.

7.3 Service Usage by Churn Status

Metric	Stayed: average	Churned: average
Seconds of Use	5,069.59	1,648.66
Frequency of SMS	84.35	16.85
Customer Value	538.59	132.18
Call Failure	7.79	7.86
Subscription Length	32.60	31.68

Customers who churned had considerably lower average call usage and SMS frequency than retained customers. Their average customer value was also substantially lower. These differences indicate that lower engagement and lower customer value were associated with churn in this dataset. Average call failure values were nearly similar for retained and churned customers, indicating no notable difference between the two groups. Retained customers had a slightly higher average subscription length than churned customers.However,the difference was relatively small.

7.4 Age Analysis

Age	Stayed	Churned	Churn rate
15	112	0	0.00%
25	764	157	17.05%
30	1,083	213	16.44%
45	293	74	20.16%
55	152	2	1.30%

Age 45 recorded the highest churn rate (20.16%) among the listed age values. Age 30 had the largest number of churned customers (213). Rate and count describe different aspects of the data, so both are useful when interpreting customer segments.

7.5 Tariff Plan Analysis

Tariff plan code	Observed churn rate
1	16.79%
2	2.62%

Tariff Plan 1 showed a higher observed churn rate than Tariff Plan 2. The dataset provides numeric plan codes, so the report does not assume names or features for either plan. The business should investigate whether pricing, included services or customer mix explain this difference.

7.6 Status Category Analysis

Status code	Observed churn rate
1	5.59%
2	47.51%

The second status code showed a considerably higher churn rate than the first. Because the precise business meaning of these codes has not been verified from a data dictionary, they are reported as codes only.

7.7 Correlation Analysis

Variable pair	Correlation	Interpretation
Frequency of SMS – Customer Value	0.92	Very strong positive correlation
Seconds of Use – Customer Value	0.41	Moderate positive correlation
Frequency of use – Customer Value	0.39	Weak-to-moderate positive correlation
Seconds of Use – Frequency of use	0.95	Very strong positive correlation
Seconds of Use – Frequency of SMS	0.09	Very weak positive correlation

Frequency of SMS had the strongest positive correlation with Customer Value among the highlighted pairs. Seconds of Use was moderately correlated with Customer Value. These correlations indicate relationships within the dataset and do not establish cause and effect.

8. Key Findings

- The overall churn rate was 15.65% (446 out of 2,850 customers).
- Customers with complaints had an observed churn rate of 82.61%, compared with 9.77% for customers without complaints.
- Churned customers had lower average call usage and SMS frequency than retained customers.
- Average Customer Value was 538.59 for retained customers and 132.18 for churned customers.
- Age 45 had the highest churn rate among the listed age values, while age 30 had the highest churn count.
- Tariff Plan code 1 had a higher observed churn rate than code 2.
- Frequency of SMS showed a very strong positive correlation with Customer Value.
- Average call failure values were nearly similar between retained and churned customers.
- Retained customers had a slightly higher average subscription length (32.60) compared to churned customers (31.68).

9. Business Recommendations

1. Improve complaint resolution: monitor complaints, reduce response delays and track whether issues are resolved satisfactorily.
2. Monitor declining usage: identify customers whose call or SMS activity falls and consider timely, relevant retention support.
3. Prioritise high-risk segments: use observed churn rates to decide which customer groups deserve further investigation, while avoiding assumptions about individuals.
4. Review tariff plans: compare plan features, pricing and customer profiles to understand the difference in churn rates between plan codes.
5. Strengthen customer engagement: consider relevant loyalty initiatives and personalised communication based on customer needs.
6. Investigate customer value: examine what drives the strong association between SMS frequency and customer value before making commercial changes.

10. Limitations

- This is an exploratory analysis and does not predict which individual customers will churn.
- Observed associations do not establish causation.
- Some variables are represented by numeric codes; their full category definitions should be confirmed using the dataset documentation.
- The findings reflect the records and variables available in this dataset and may not generalise to every telecom company or market.
- No statistical significance testing or controlled causal analysis is reported in this version.

11. Conclusion

This project used Python to explore customer churn in a telecommunications dataset. After removing duplicate records, 2,850 customer records were analysed. The overall churn rate was 15.65%. The analysis identified notable differences in churn across complaint status, service usage, customer value, age values and tariff-plan codes.

The strongest observed pattern was the high churn rate among customers who had recorded complaints. Churned customers also had lower average call usage, SMS frequency and customer value than retained customers. These findings can support further investigation and help a telecom business focus on complaint resolution and customer engagement. The results should be interpreted as exploratory associations rather than causal conclusions.

Appendix: Analysis Environment

The detailed code, cell outputs and visualisations are maintained in the supporting Jupyter Notebook. This report summarises the analysis results available at the time of preparation. The notebook and report should be checked together before final submission.